UNIT 2 · TOPIC 6(A)

Design of Seed Metering Systems

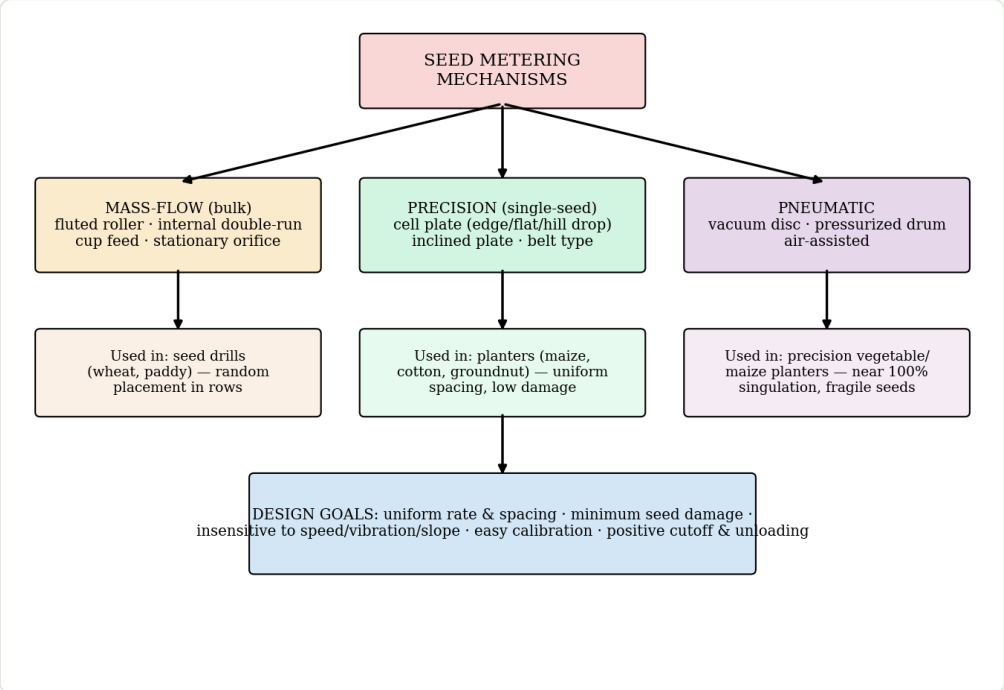
Syllabus topic: Design of soil working tools for sowing and planting

Functional requirements of a sowing/planting machine

1. Meter seeds at the desired rate with uniform distribution/spacing.
2. Open a furrow to uniform depth.
3. Deposit the seed with minimum bounce/scatter.
4. Cover and firm the soil around the seed.
5. Cause minimum mechanical damage to the seed.

Kepner's requirements of a good precision metering unit (§11.16 area):

- Cells sized to the seed (one seed per cell — grading the seed helps).
- Proper cell fill (**95–100%** target).
- Gentle handling (low plate speed, smooth surfaces).
- A good **cutoff device** (prevents multiple fill and seed breakage).
- **Positive unloading** — ejection of seed from cells (knock-out devices).

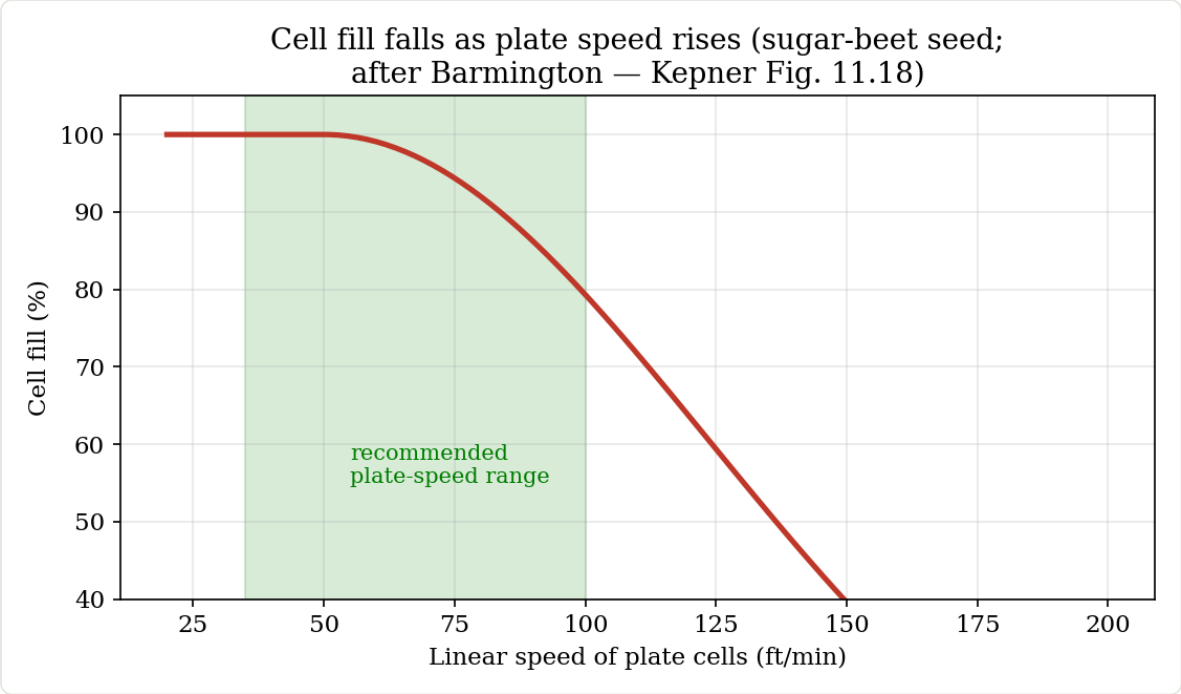


Classification of seed metering mechanisms

Cell-plate (precision) metering — design variables

- **Cell size vs seed size:** make the cell slightly larger than the largest seed dimension. Graded/processed seed (e.g., segmented sugar-beet seed, delinted cotton) enables true singulation.
- **Plate (cell) linear speed** is the critical design limit — cell fill falls as plate speed rises. Test data (Barmington, in Kepner Fig. 11.18): **100% cell fill at ~50 fpm (~0.25 m/s)**; good range ≈ **35–100 fpm**. 95–100% cell fill also gives the most uniform spacing and minimum seed damage.
- 100% "fill" is statistical — empty cells are offset by multiple fills.

- For higher forward speeds, use **more cells per plate or a larger plate diameter — not a faster plate**.



Cell fill vs plate speed

$$s = \pi \cdot D_w(1 - \sigma)/(n_c \times i)$$

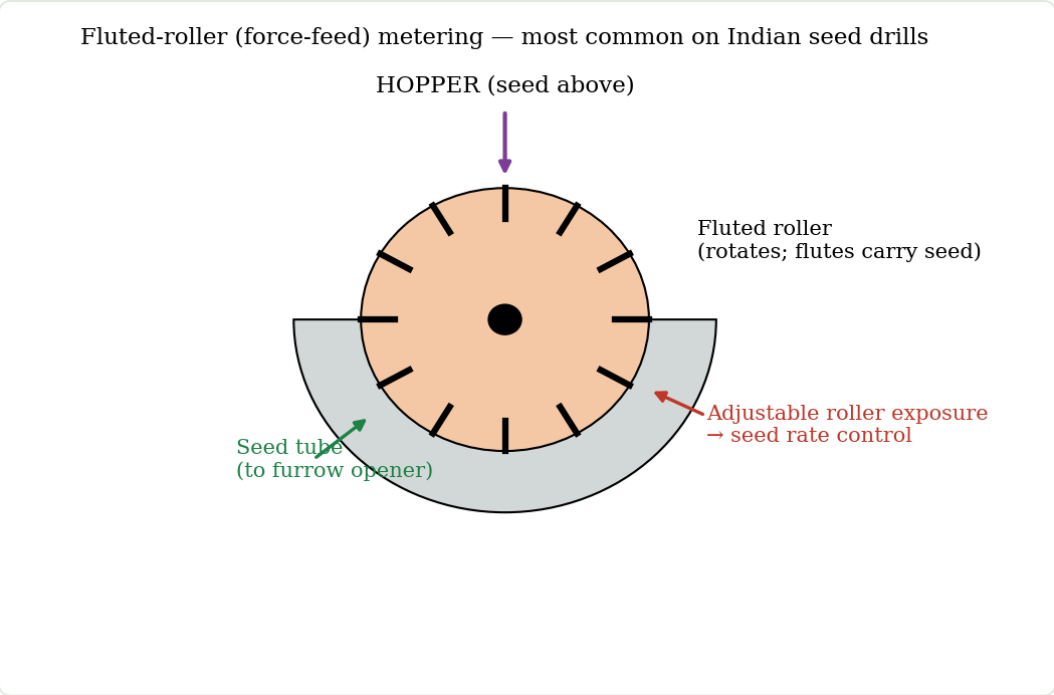
Seed spacing delivered by a planter: D_w = drive-wheel diameter, σ = wheel slip (5–10%), n_c = cells per plate, i = plate-to-wheel speed ratio.

$$\text{Plant population/ha} = 10^4/(s \times \text{row spacing})$$

Sizing starts from the agronomic population.

- **Vacuum disc design:** orifice dia ≈ **0.4–0.6 × seed dia**; vacuum **30–60 mbar** (higher for heavier seed); a singulator (scraper) removes doubles.

Mass-flow metering (drills)

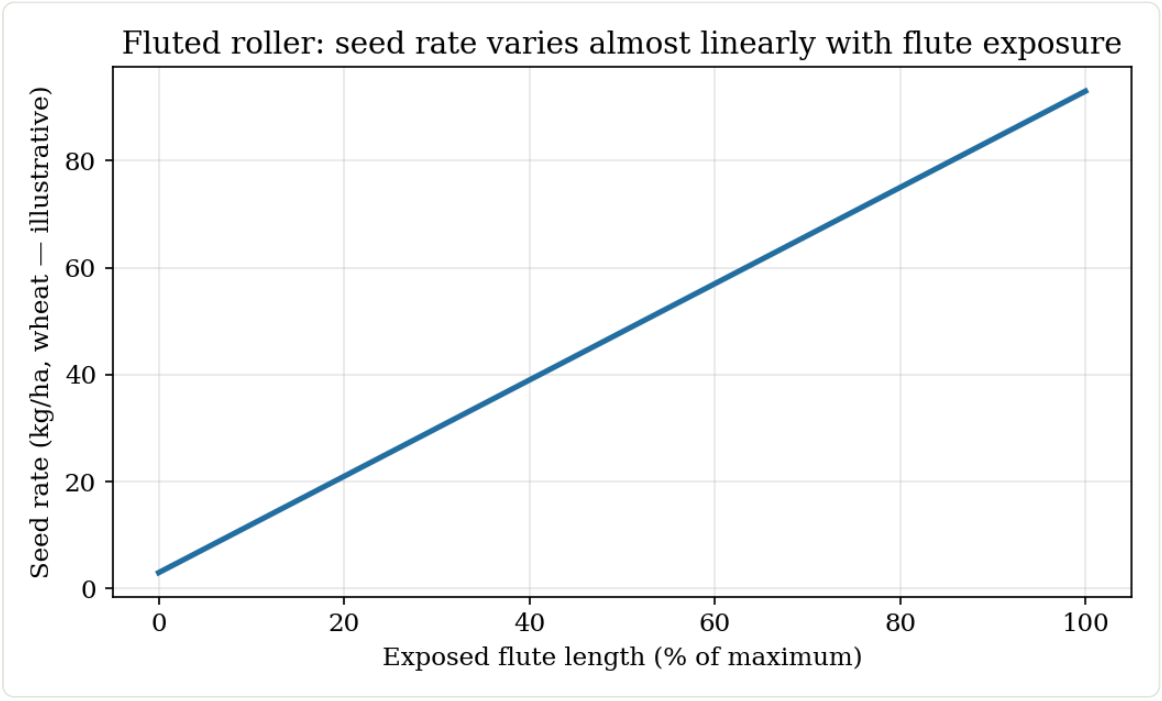


Fluted roller metering

- **Fluted roller (force feed):** seed rate ∝ exposed flute length (near-linear) and roller rpm. Size the flute shape (straight/helical — helical gives steadier flow), roller diameter and groove radius so seeds are not sheared at the cutoff; an adjustable spring-loaded gate protects large seeds.

$$Q\ (\mathrm{kg/s}) = \rho_b \times (a_f \times L_e \times z \times N/60) \times \eta_f$$

Bulk density × flute cross-section × exposed length × number of flutes × rpm × fill factor.



Seed rate vs flute exposure

- **Internal double-run:** two-diameter compartments for coarse/fine seeds; set the rate by speed ratio (gearing) rather than exposure.

- **Drive design:** ground wheel → chain/gear ratios; change the rate in steps with the gear ratio and finely with flute exposure.

Practice MCQs

Q111. The target cell fill for a precision planter plate is:

(a) 50–60% (b) 70–80%
(c) 95–100% (d) Exactly 110%

Q112. As plate (cell) linear speed increases, cell fill:

(a) Increases (b) Decreases
(c) Constant (d) First falls then rises

Q113. Barmington's data (Kepner Fig. 11.18) showed 100% cell fill for sugar-beet seed at a plate speed of about:

(a) 10 fpm (b) 50 fpm
(c) 300 fpm (d) 500 fpm

Q114. To plant at higher forward speeds with a plate meter, the designer should:

(a) Speed up the plate (b) Use more cells per plate or a larger plate
(c) Reduce cells (d) Raise the hopper

Q115. The metering-unit element that prevents multiple fill and seed breakage is the:

(a) Knock-out (b) Cutoff device
(c) Agitator (d) Gate spring

Q116. Positive removal of seed from cells is ensured by:

(a) Gravity only (b) Knock-out/ejector devices
(c) Vacuum (d) Higher speed

Q117. Conditions giving 95–100% cell fill also give:

(a) Max damage (b) Most uniform spacing and minimum seed damage
(c) Higher draft (d) Poor spacing

Q118. In fluted-roller metering, seed rate is chiefly controlled by:

(a) Roller diameter (b) Exposed flute length
(c) Seed tube length (d) Hopper level

Q119. Helical flutes (vs straight) on a fluted roller give:

(a) Higher damage (b) Steadier, more uniform seed flow
(c) Lower capacity only (d) More wear

Q120. Seed spacing delivered by a planter (D_w wheel dia, σ slip, n_c cells, i plate/wheel ratio) is:

(a) $\pi D_w(1-\sigma)/(n_c \cdot i)$ (b) $n_c \cdot i/\pi D_w$
(c) $\pi D_w \cdot n_c \cdot \sigma$ (d) D_w/n_c

Q121. Ground-wheel slip (skid) of a drill causes actual seed spacing to be:

(a) Narrower than theoretical (b) Wider than theoretical
(c) Unchanged (d) Random

Answers

Q111 — (c) 95–100%
Q112 — (b) Decreases
Q113 — (b) 50 fpm

Q114 — (b) Use more cells per plate or a larger plate
Q115 — (b) Cutoff device
Q116 — (b) Knock-out/ejector devices
Q117 — (b) Most uniform spacing and minimum seed damage
Q118 — (b) Exposed flute length
Q119 — (b) Steadier, more uniform seed flow
Q120 — (a) $\pi D_w(1-\sigma)/(n_c \cdot i)$
Q121 — (b) Wider than theoretical

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 11 (Crop Planting), Ch. 12 (Row-Crop Cultivation), Ch. 13 (Application of Fertilizers); standard FMP design references.